INCH-POUND

MIL-S-15291/11A(SH)
22 July 1991
SUPERCEDING
MIL-S-15291/11(SH)
17 April 1981

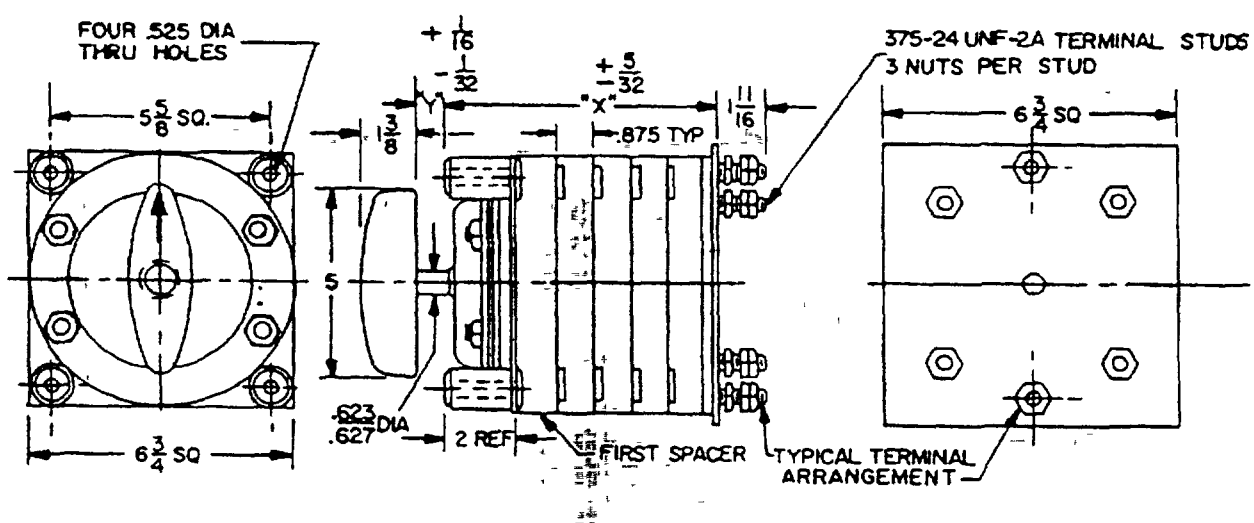
MILITARY SPECIFICATION SHEET

SWITCHES, ROTARY, SNAP ACTION CLASS 20SR FRONT MOUNTED, BACK CONNECTED

Inactive for new design after 17 April 1981.
For new design use MIL-S-15291/12(SH)

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-S-15291.



NOTES
1 Dimensions are in inches
2 Unless otherwise specified, tolerance is plus or minus 0.015

FIGURE 1 *Class 20SR switch (up to 6 studs).*

AMSC N/A FSC 5930

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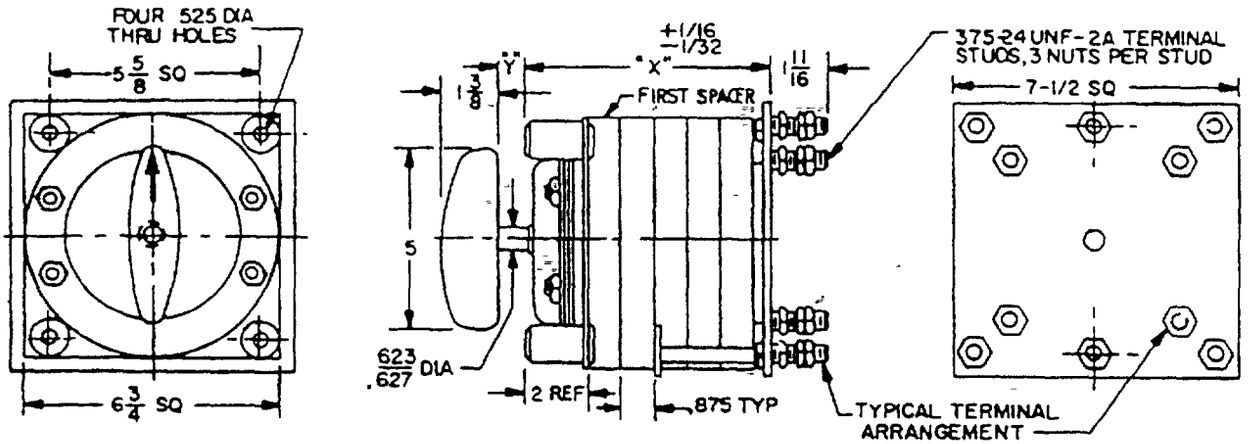


FIGURE 2 *Class 20SR switch (up to 10 studs)*

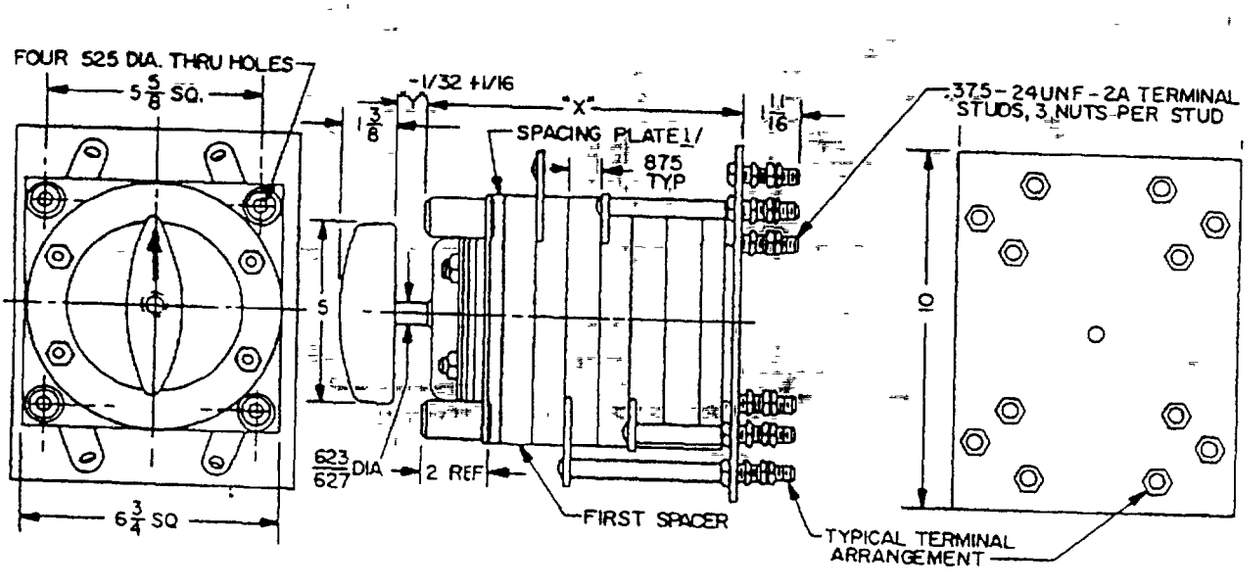
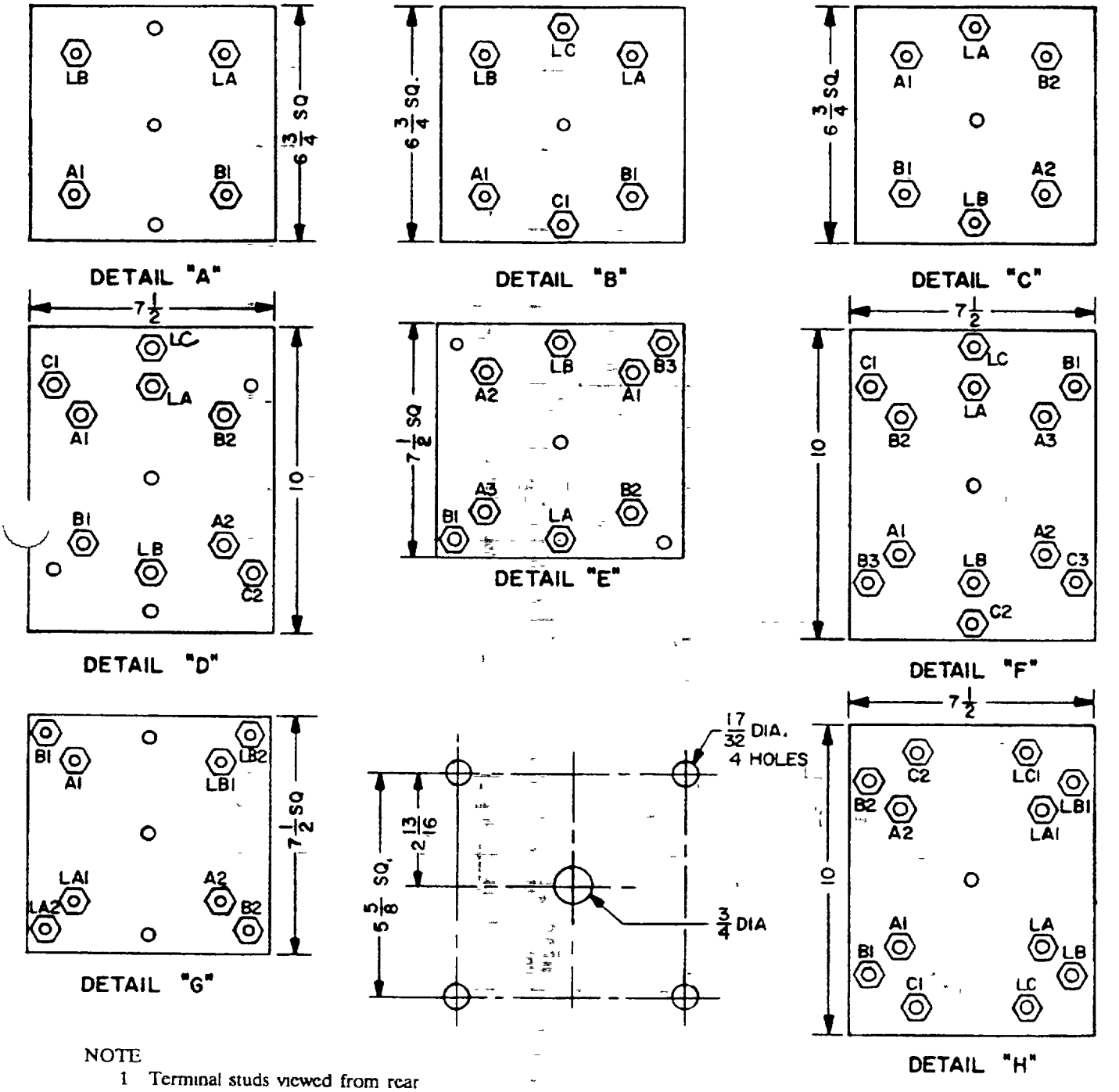


FIGURE 3 *Class 20SR swtch (up to 12 studs)*

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NOTE
1 Terminal studs viewed from rear

FIGURE 4 *Mounting dimensions and terminal stud locations*

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TABLE I *Type and switching characteristics.*

MIL-S-15291/11 Dash No Type designation	Detail ref / Torque in-lbs	Dim "X" / Dim "Y"	Handle or shaft position	Circuit and spacer configuration rotor position, spacer locations, terminal marking and locations: No 1	No 2	No 3	No 4	No 5	No 6	No 7	No 8	No 9	Notes
-001 20SR2A1	Figs 1 & 4 Detail A 72	4 15/16 7/16	OFF ON OFF		LB B1	LA A1							
-002 20SR3A1	Figs 1 & 4 Detail B 72	5 13/16 7/16	OFF ON OFF		LC C1	LB B1	LA A1						
-003 20SR2B1	Figs 1 & 4 Detail C 72	6 11/16 7/16	OFF ON 1 ON 2		LB B2	B2 B1	LA A2	A1 A2					
-004 20SR3B1	Figs 3 & 4 Detail D 72	8 7/16 7/16	OFF ON 1 ON 2		LC C2	C1 C2	LB B1	B2 B1	LA A2	A1 A2			
-005 20SR2E1	Figs 2 & 4 Detail E 72	6 11/16 7/16	OFF ON 1 ON 2		LB B3	B3 B2	LA A3	A1 A2					
-006 20SR3E1	Figs 3 & 4 Detail F 72	8 7/16 7/16	OFF ON 1 ON 2		LC C3	C1 C2	LB B3	B2 B3	LA A3	A3 A1			
-007 20SR4F1	Figs 2 & 4 Detail G 72	6 11/16 7/16	OFF ON 1 ON 1&2		LB2 B2	LB1 B1	A2 LA2 B2	A1 LA1					
-008 20SR6F1	Figs 3 & 4 Detail H 80	8 7/16 7/16	OFF ON 1 ON 1&2		LC1 C2 LC	LB1 B2 C1	LB B1	LA1 A2 LA					

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REQUIREMENTS:

Applicable specification:	MIL-S-15291
Dimensions and mounting.	See figures 1, 2, 3, and 4
Switching characteristics	See table I
Angle of throw:	90 degrees
Switching action:	Snap action, reciprocating
Electrical and endurance ratings:	See table II
Stop strength (applicable to switches with stops):	100 inch-pounds
Vibration:	50 Hz, MIL-STD-167-1
Shock:	High impact, MIL-S-901
Insulation (spacers and decks)	Glass alkyd, MAI-60 (or MAI-30) in accordance with MIL-M-14
Spacer plate:	MAI-60, MAI-30 of GFI-10 in accordance with MIL-M-14
Contact resistance:	1.5 milliohms maximum
Dielectric withstanding voltage:	2,000 Vrms
Insulation resistance:	200 megohms minimum
Temperature rise:	50 °C maximum
Contact material:	Rotor, phosphor bronze – Stationary contact, copper
Mounting bolts:	(4) 1/2 diameter, length, and headstyle to suit application. Bolts not furnished.
Handle:	M15291/12-100.

TABLE II *Electrical and endurance ratings.*

Tests	Current (amperes)	Voltage (volts)	Electrical operations (number of operations)	Test rate (operations per minute)	De-energized operations (number of operations)
Alternating current (rms)	200	500	6,000	8	4,000
Direct current	200	250	4,000	3	-

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Terminal marking: Terminal markings shall be in accordance with details A through H, as specified in table I.

GENERAL INFORMATION:

Switches not covered by specification sheets: Switches which are fabricated from standard parts, as used in qualified switches, but which do not comply with switches detailed herein with respect to circuit characteristics, switching action, mounting arrangement, and handle details may be acquired under this specification from contractors having qualification approval under this specification.

Extended ratings: Switches detailed herein have been tested and found satisfactory at the extended ratings listed in table III. Reduced life expectancy must be anticipated for switches used at these increased voltage or current levels. Tests under the conditions of table III are not required for qualification acceptance and they are not repeated routinely as for maintenance of qualification Supplemental evaluations and tests applicable to particular circuit requirements are recommended.

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TABLE III. *Extended ratings.*

Switching characteristics	AC - 60 or 400 Hz											
	125 volts				250 volts				500 volts			
	Resistive or lamp load		Inductive load 0.75 p.f.		Resistive or lamp load		Inductive load 0.75 p.f.		Resistive or lamp load		Inductive load 0.75 p.f.	
	Amp.	Operations	Amp.	Operations	Amp.	Operations	Amp	Operations	Amp.	Operations	Amp.	Operations
All	200	6,000	200	6,000	200	6,000	200	6,000	200	6,000	200	6,000
	DC											
	120 volts				250 volts				350 volts			
	Resistive or lamp load		Inductive load		Resistive or lamp load		Inductive load [1]		Resistive or lamp load		Inductive load	
	Amp.	Operations	Amp.	Operations	Amp.	Operations	Amp.	Operations	Amp.	Operations	Amp.	Operations
All	200	4,000	---	---	200	4,000	200	4,000	---	---	---	---

[1] 0.025 henry for inductive circuit

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APPLICATION AND ACQUISITION GUIDE: PIN and type designation cross reference shall be as shown in table IV.

TABLE IV. *Application and acquisition guide*

M15291/11 dash No	Type designation	For replacement only In new design use	Circuit configuration
-001	20SR2A1	M15291/12-001	A (off-on-off-on)
-002	20SR3A1	M15291/12-002	A (off-on-off-on)
-003	20SR2B1	M15291/12-003	B (off-on 1-off-on 2)
-004	20SR3B1	M15291/12-004	B (off-on 1-off-on 2)
-005	20SR2E1	M15291/12-005	E (off-on 1-on 2-on 3)
-006	20SR3E1	M12591/12-006	E (off-on 1-on 2-on 3)
-007	20SR4F1	M12591/12-007	F (off-on 1-on 1 & 2-on 2)
-008	20SR6F1	M12591/12-008	F (off-on 1-on 1 & 2-on 2)

Revision letters are not used to denote changes due to the extensiveness of the changes

Preparing activity:
Navy - SH
(Project 5930-N663-12)

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